

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)*

Assessment Tool Weightage: 20%

1. Construct a comprehensive and well-organized concept map that accurately identifies all key concepts from literature involved in the pipeline from realworld scene capture to digital image representation, establishes clear and meaningful relationships among intermediate transformations and dependencies, integrates insights from multiple studies, and presents the information in a clear and visually structured manner

Ans)

Real-World Scene



Image Capture



Sampling



Quantization



Digital Image



Image Processing



Computer Vision



Applications

2. Develop a hierarchically structured concept map that identifies major stages of visual data handling in a Computer Vision system, clearly illustrates logical relationships and contributions of each stage toward generating meaningful outputs, incorporates relevant literature connections, and ensures clarity and readability.⁶⁶ concepts related to image resolution and intensity levels, establishes clear cause–effect relationships with image quality and usability, integrates supporting evidence from literature, and presents the concepts in a logically organized and visually clear format.

ANS)

Computer Vision System

Visual Data Input



Image Acquisition



Image Preprocessing



Feature Extraction



Image Analysis



Object Detection



Classification



Meaningful Output

3. Design an analytically organized concept map that identifies key factors in image acquisition conditions, clearly links them to downstream processing and application performance through meaningful relationships, integrates findings from relevant studies, and ensures a well-structured and easy-to-interpret presentation.

Image Acquisition Conditions

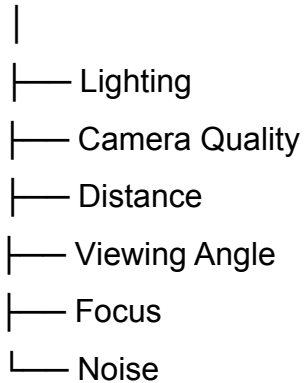


Image Quality



Image Preprocessing



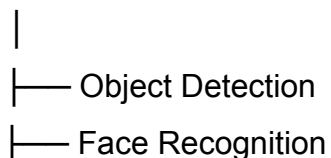
Feature Extraction

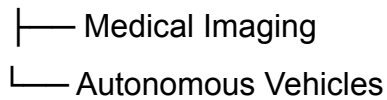


Image Analysis



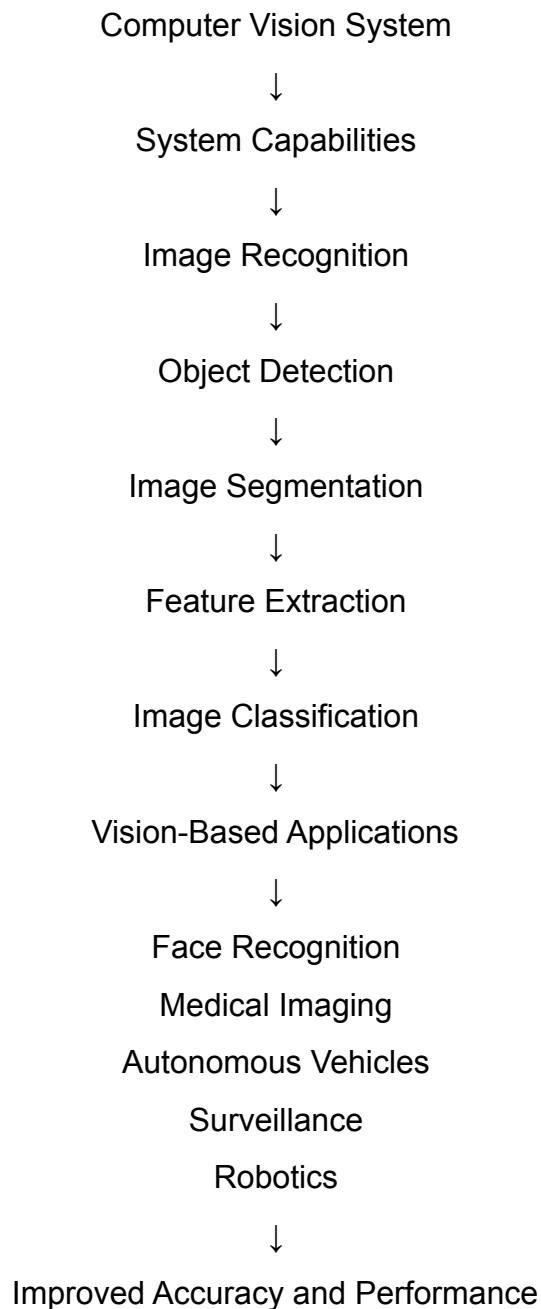
Application Performance



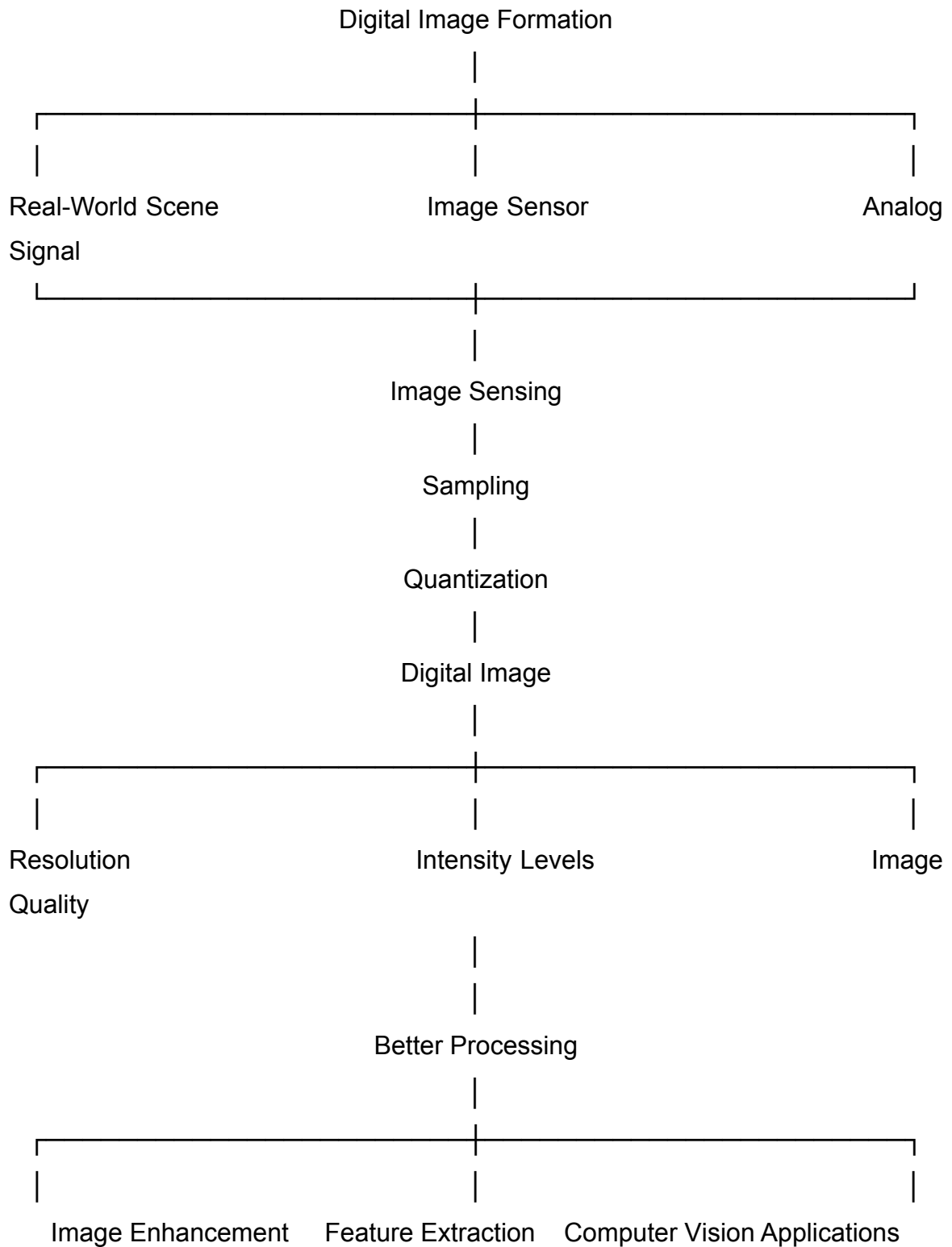


4. Construct an original and logically structured concept map that identifies system capabilities and their influence on the selection and effectiveness of vision-based applications, establishes clear interconnections, integrates multiple literature perspectives, and maintains high clarity and visual organization.

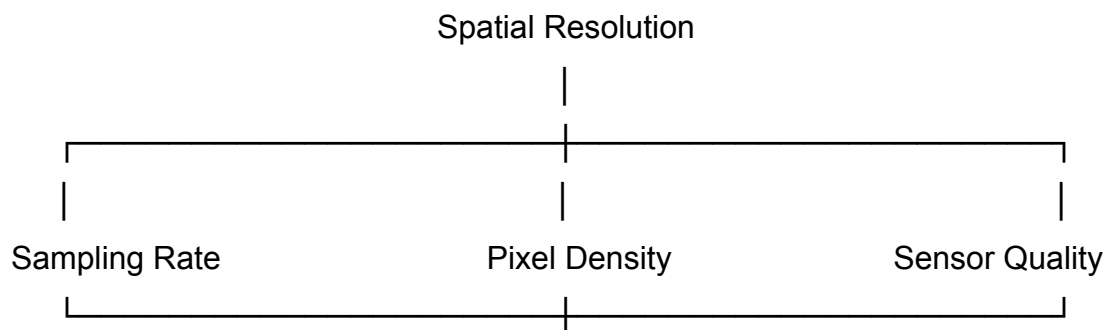
ANS)

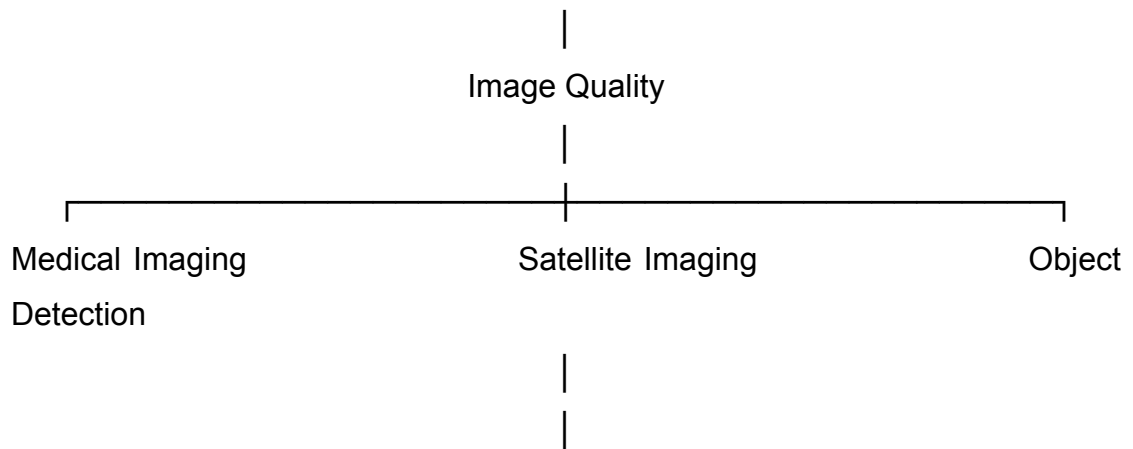


5. Construct a concept map that identifies the complete workflow of digital image formation, clearly linking sensing, sampling, and quantization stages with their impact on image quality, integrating relevant literature insights, and presenting the relationships in a structured and visually coherent manner.

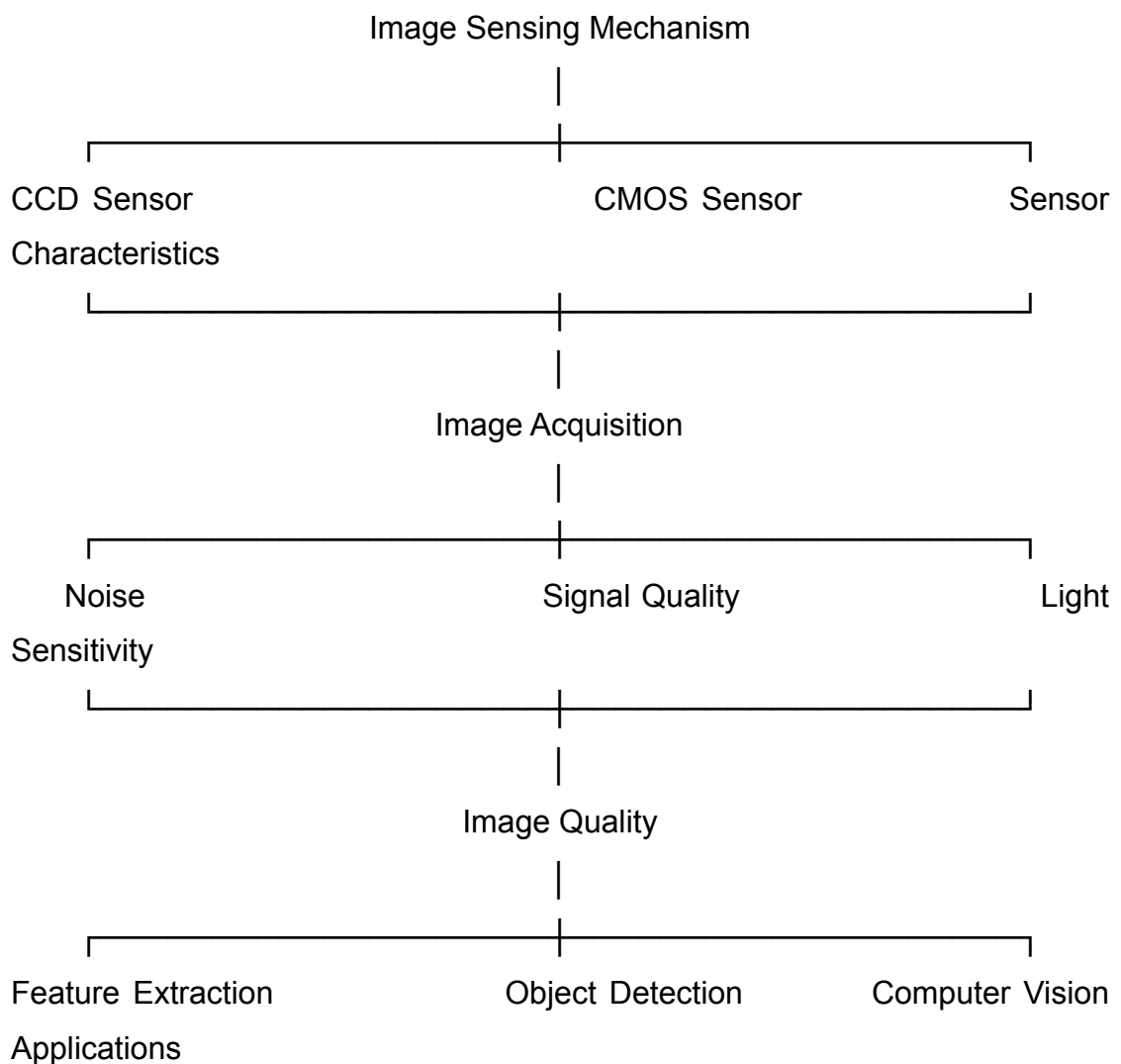


7. Create a concept map that identifies factors affecting spatial resolution, clearly linking them to sampling rate, pixel density, and application requirements, incorporating supporting evidence from literature, and presenting the structure with clarity.

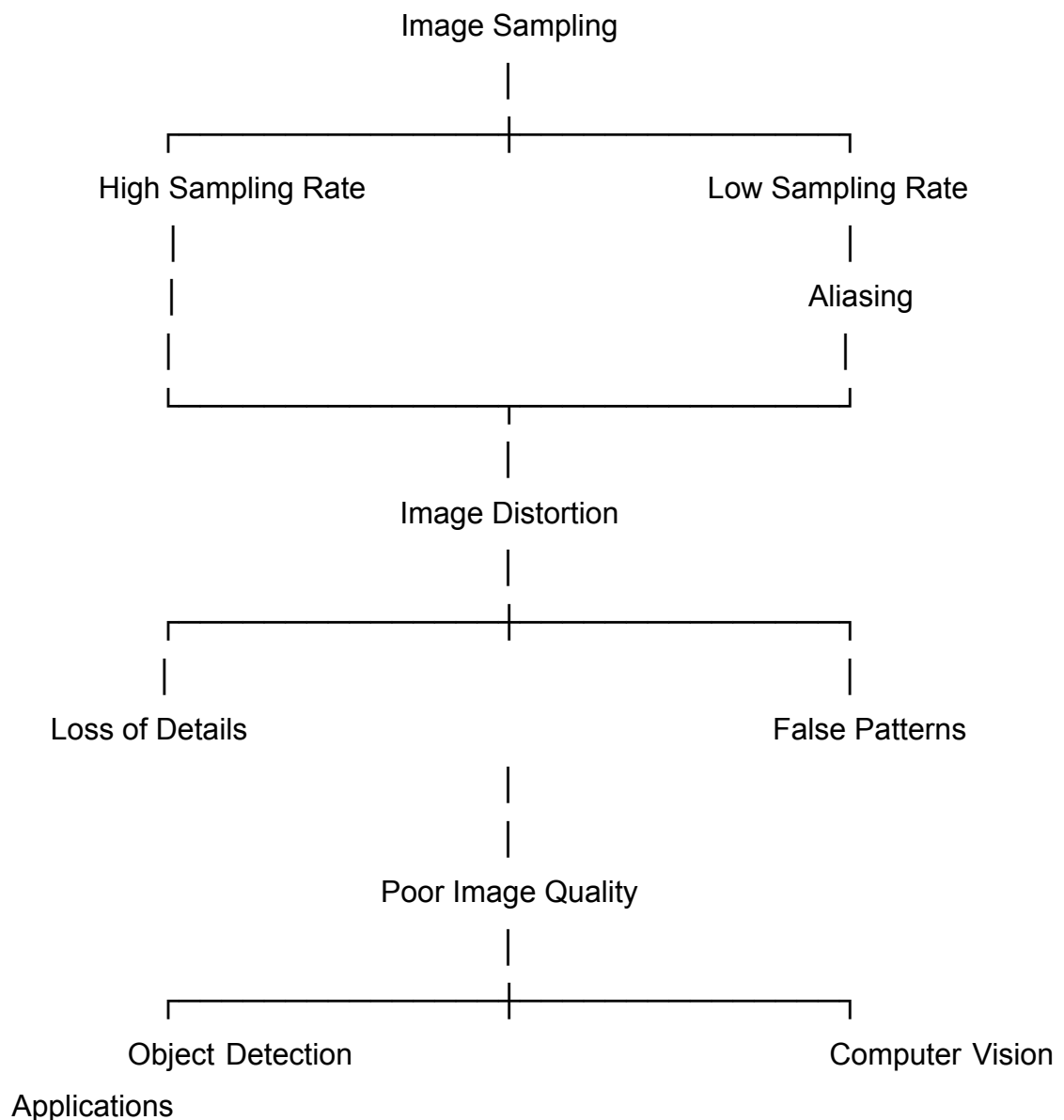




8. Design a concept map that identifies the role of image sensing mechanisms, clearly establishing relationships between sensor characteristics, noise generation, and acquisition quality, integrating literature perspectives, and ensuring logical flow.

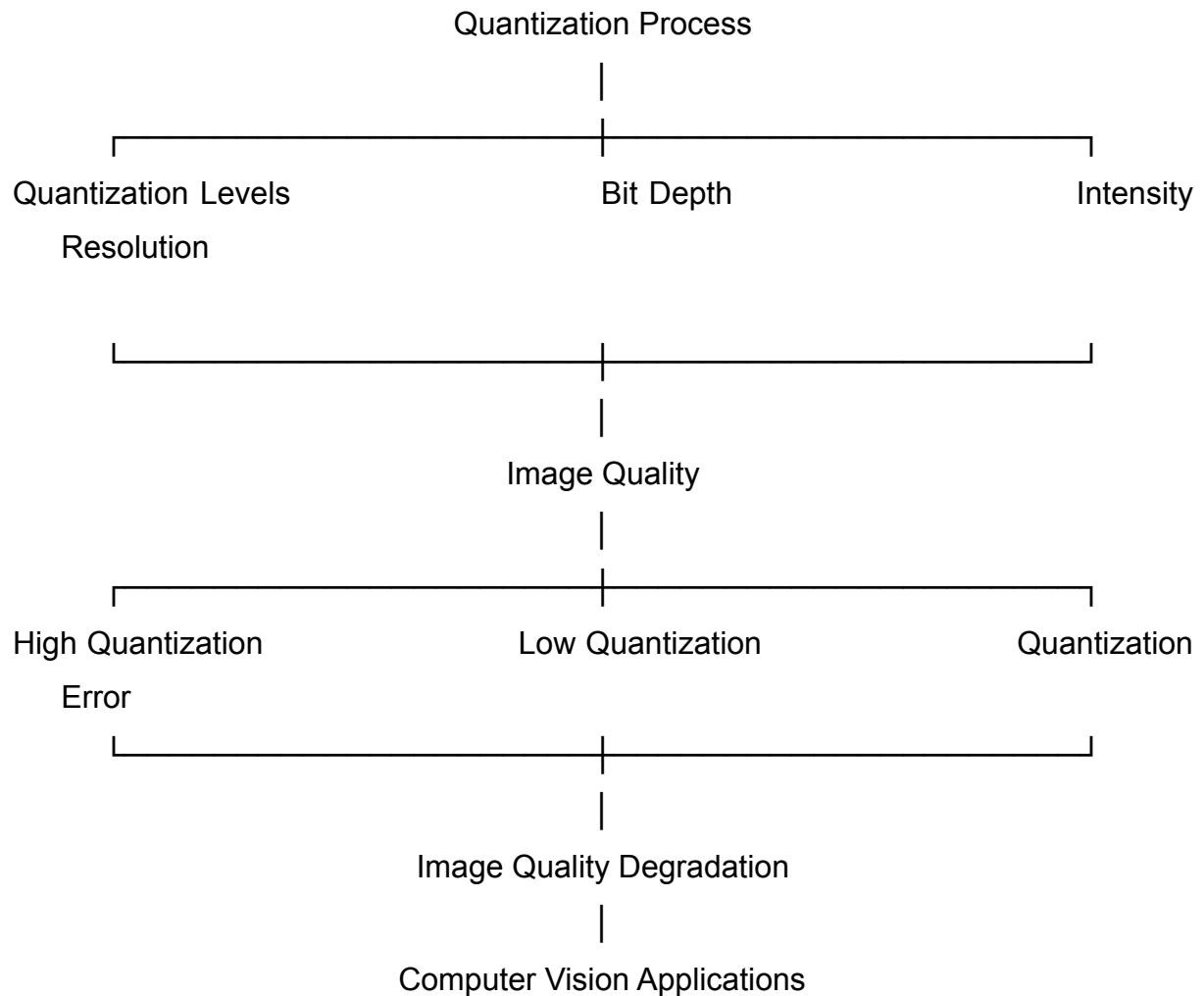


10. Construct a concept map that identifies the relationship between sampling, aliasing, and image distortion, clearly illustrating cause–effect dependencies, integrating relevant theoretical insights, and presenting the structure in a visually organized manner.

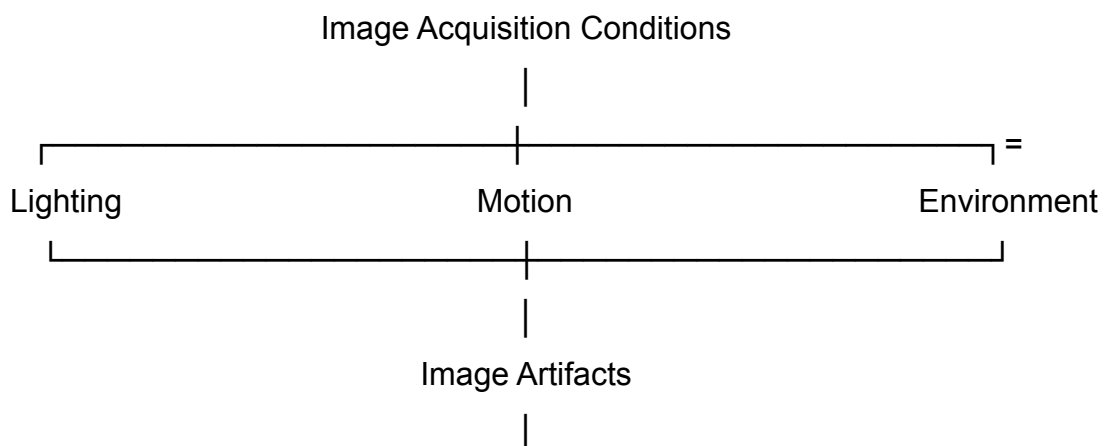


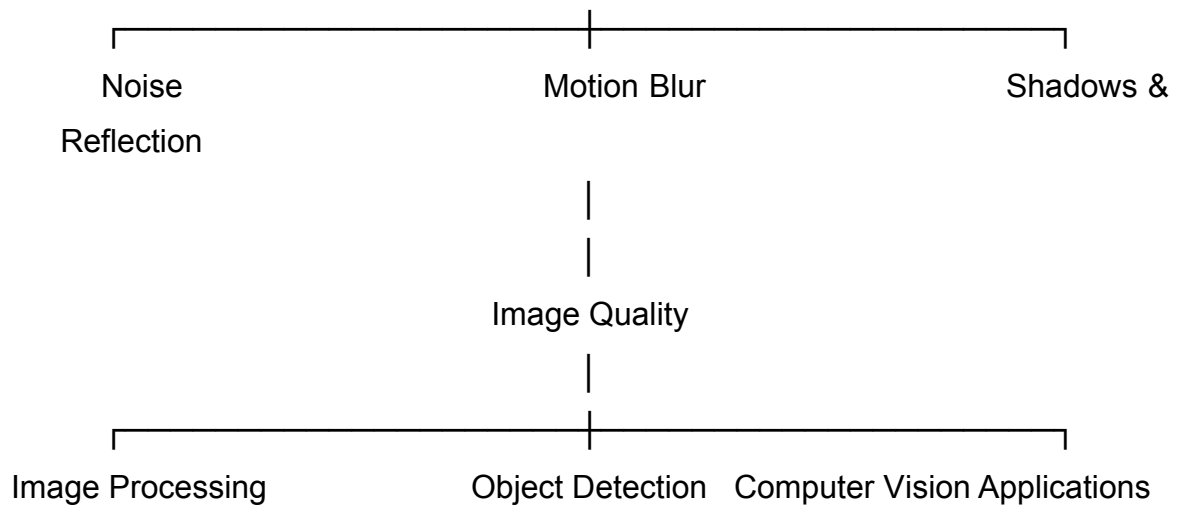
11. Develop a concept map that identifies quantization processes and their impact on intensity resolution, clearly linking quantization levels to image quality

degradation, incorporating literature references, and ensuring structured visualization.

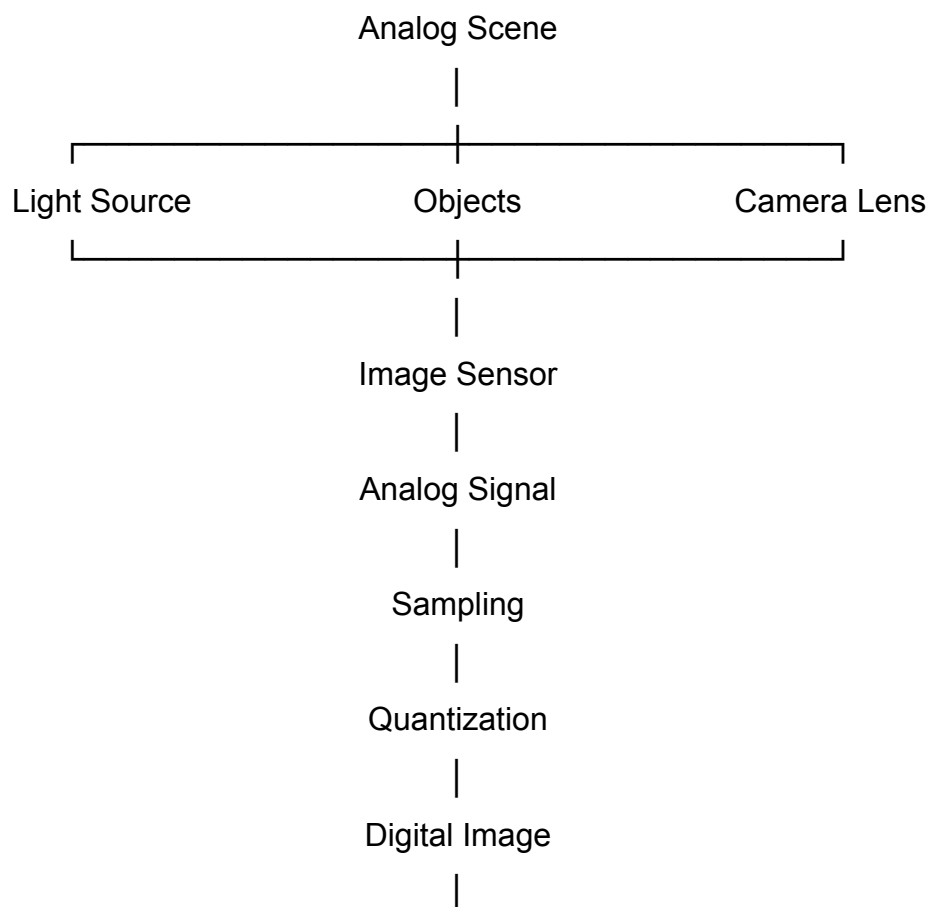


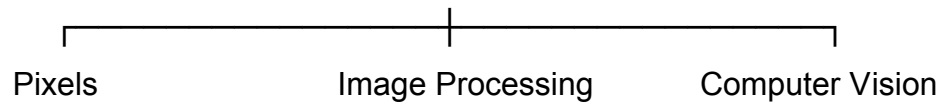
12. Create a concept map that identifies the relationship between image acquisition conditions (lighting, motion, environment) and resulting image artifacts, integrating findings from literature and presenting clear interconnections.



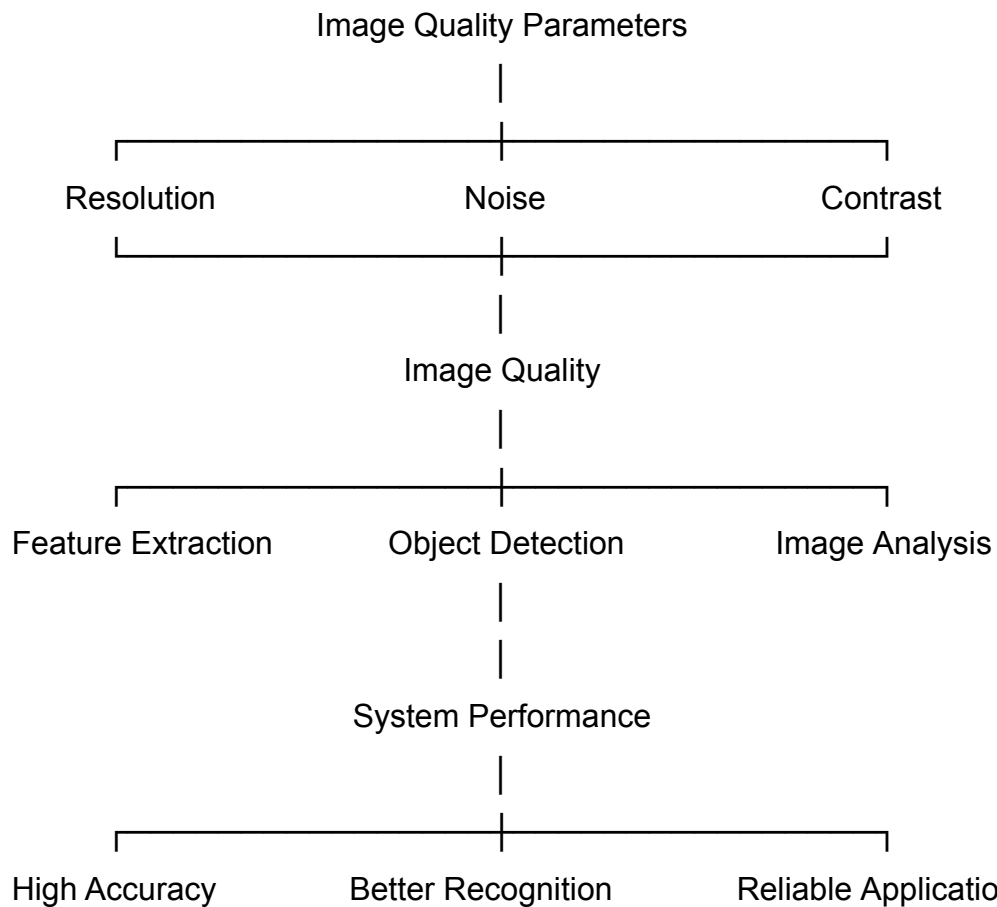


13.Design a concept map that identifies the transformation from analog scene to digital image, clearly linking physical processes to computational representation, integrating literature-based insights, and ensuring clarity and organization.

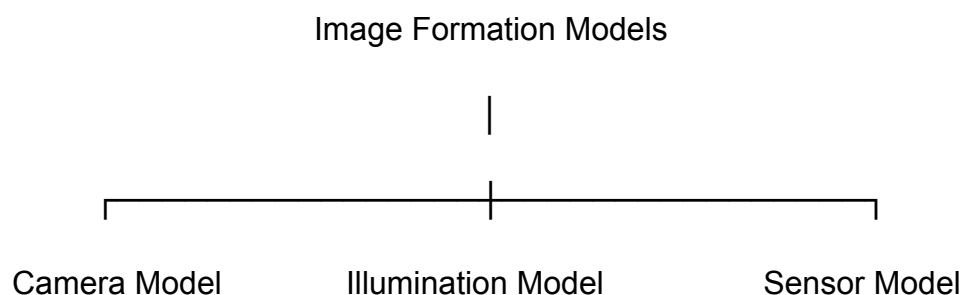


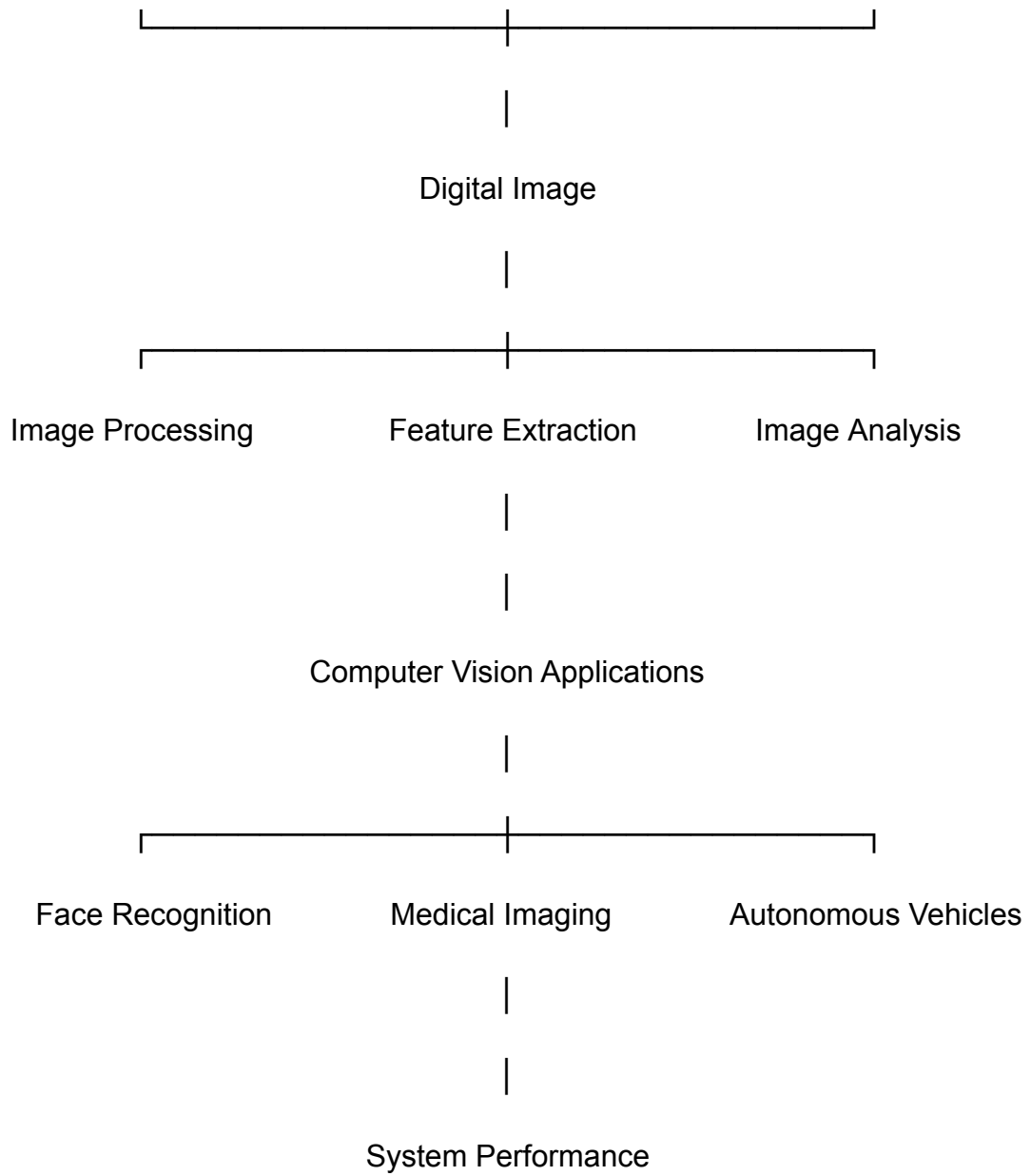


14. Construct a concept map that identifies dependencies between image quality parameters (resolution, noise, contrast) and system performance, integrating multiple perspectives from literature and presenting structured relationships.



15. Develop a concept map that identifies interactions between image formation models and Computer Vision applications, clearly linking theoretical models to practical system performance, integrating literature insights, and ensuring





Code of Conduct:

I K.L.V.S Karthik(192424177) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.L.V.S Karthik

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately	Identifies most key concepts with minor	Identifies limited concepts; some are	Fails to identify essential concepts

		and comprehensively	omissions	irrelevant or missing	
Linkages	25%	Establishes clear, meaningful, and accurate relationships	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially	No meaningful relationships established

		between concepts		incorrect	
Organization	20%	Concepts are wellstructured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration ; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand